

Part-II Relativity — Master Cheat Sheet

Handouts + Example Sheets (signature + - - -; $\mu \equiv GM/c^2$; $\kappa = 8\pi G/c^4$; items tagged (ex. N QM) come from Example Sheet N, Question M)

1. SPECIAL RELATIVITY

Postulates: laws same in all inertial frames; c invariant. Frames differ by translation, rotation, constant-velocity boost.

Lorentz transformations

Standard config (boost $v = \beta c$ along x), $\gamma = (1 - \beta^2)^{-1/2}$:

$$ct' = \gamma(ct - \beta x), \quad x' = \gamma(x - \beta ct)$$

$$y' = y, \quad z' = z$$

Inverse: $v \rightarrow -v$. **Rapidity** $\beta = \tanh \psi$, $\gamma = \cosh \psi$, $\gamma\beta = \sinh \psi$; a boost is a hyperbolic rotation in $ct-x$; colinear boosts: rapidities add.

General boost (ex.1 Q4), $\vec{\beta} = \vec{v}/c$, $\alpha = (\gamma - 1)/|\vec{\beta}|^2$:

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta_j \\ -\gamma\beta_i & \delta_{ij} + \alpha\beta_i\beta_j \end{pmatrix}$$

(resolve $\vec{r}$ into parts $\parallel, \perp$ to $\vec{\beta}$).

Interval & causal structure

Invariant interval $\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$. Sign is invariant: > 0 timelike, $= 0$ null, < 0 spacelike. Causal order of timelike/null-separated events is invariant; simultaneity is not.

Kinematic effects

Length contr. $\ell = \ell_0/\gamma$ ($\parallel v$ only); $V = V_0/\gamma$. **Time dilation** $d\tau = (1 - v^2/c^2)^{1/2}dt = dt/\gamma$. **Velocity addition** $u'_x = \frac{u_x - v}{1 - u_x v/c^2}$, $u'_{y,z} = \frac{u_{y,z}}{\gamma(1 - u_x v/c^2)}$. **Doppler** $\frac{\lambda}{\lambda_0} = \gamma(1 - \beta \cos \theta)$; longitudinal $\sqrt{\frac{1 - \beta}{1 + \beta}}$. **Aber-**

ration (ex.1 Q6): transform $\tan \theta$ via velocity components; for a photon use $u' = c$.

Acceleration

Not invariant but absolute. Rectilinear, proper accel $f(\tau)$:

$$\frac{du}{d\tau} = \left(1 - \frac{u^2}{c^2}\right)f, \quad c\psi(\tau) = \int_0^\tau f d\tau'$$

Const f : $\psi = f\tau/c$, $ct = (c^2/f) \sinh \psi$, $x = (c^2/f)(\cosh \psi - 1)$ (hyperbola; gives an event horizon for the accelerated observer).

2. MANIFOLDS & METRIC

Line element $ds^2 = g_{ab}(x) dx^a dx^b$, $g_{ab} = g_{ba}$; $N(N+1)/2$ indep. functions. **Transformation** $g'_{ab} = \frac{\partial x^c}{\partial x'^a} \frac{\partial x^d}{\partial x'^b} g_{cd}$. **Volume element** $dV = \sqrt{|g|} dx^1 \cdots dx^N$ (invariant; $g' = g/J^2$). **Curve length** $L = \int |g_{ab} \dot{x}^a \dot{x}^b|^{1/2} du$.

Standard examples

2-sphere (radius a): $ds^2 = a^2(d\theta^2 + \sin^2 \theta d\phi^2)$.

3-sphere (ex.1 Q10):

$$ds^2 = a^2[d\chi^2 + \sin^2 \chi(d\theta^2 + \sin^2 \theta d\phi^2)]$$

Area of 2-sphere χ_0 : $4\pi a^2 \sin^2 \chi_0$; total volume $2\pi^2 a^3$.

3. VECTORS & TENSORS

Sym/antisym $S_{(ab)} = \frac{1}{2}(S_{ab} + S_{ba})$, $S_{[ab]} = \frac{1}{2}(S_{ab} - S_{ba})$; general $S_{[ab \cdots c]} = \frac{1}{n!}$ alternating sum.

Contraction identities (ex.2 Q2) $A^{ab}T_{ab} = A^{ab}T_{[ab]}$, $S^{ab}T_{ab} = S^{ab}T_{(ab)}$.

Raise/lower $v_a = g_{ab}v^b$, $v^a = g^{ab}v_b$; $g^{ab}g_{bc} = \delta^a_c$. **Quotient theorem** establishes tensor character from a contraction. **Curl test** $A_{ab} = \partial_b v_a - \partial_a v_b$ is a $(0,2)$ tensor; $B_{abc} = \partial_c A_{ab} + \partial_a A_{bc} + \partial_b A_{ca}$ a totally-antisym $(0,3)$ tensor.

4. TENSOR CALCULUS

Covariant derivative

$$\nabla_a \phi = \partial_a \phi;$$

$$\nabla_a v^b = \partial_a v^b + \Gamma_{ac}^b v^c, \quad \nabla_a X_b = \partial_a X_b - \Gamma_{ab}^c X_c$$

Metric connection (torsion-free, $\nabla_a g_{bc} = 0$):

$$\Gamma_{bc}^a = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc})$$

Diagonal-metric shortcut (ex.2 Q3; $a \neq b \neq c$, no sum):

$$\Gamma_{bc}^a = 0, \quad \Gamma_{aa}^b = -\frac{1}{2g_{bb}}\partial_b g_{aa}, \quad \Gamma_{ba}^a = \partial_b \ln \sqrt{|g|}$$

Also $\partial_c g = g g^{ab} \partial_c g_{ab}$.

Vector operators

Divergence $\nabla_a v^a = |g|^{-1/2} \partial_a (|g|^{1/2} v^a)$. **Laplacian** $\nabla^2 \phi = |g|^{-1/2} \partial_a (|g|^{1/2} g^{ab} \partial_b \phi)$. **Curl** $(\text{curl } X)_{ab} = \partial_a X_b - \partial_b X_a$ (connection-free). **Plane polar** (ex.2 Q4): $\Gamma_{\rho\phi}^\phi = 1/\rho$, $\Gamma_{\phi\phi}^\rho = -\rho$; $\nabla_a v^a = \frac{1}{\rho} \partial_\rho (\rho v^\rho) + \partial_\phi v^\phi$.

Transport along a curve

Intrinsic deriv. $\frac{Dv^a}{Du} = \frac{dx^b}{du} \nabla_b v^a = \frac{dv^a}{du} + \Gamma_{bc}^a \frac{dx^b}{du} v^c$. **Parallel transport** $Dv^a/Du = 0$ (so $\delta v^a = -\Gamma_{bc}^a \delta x^b v^c$); preserves lengths & scalar products; path-dependent iff curved.

5. GEODESICS

Affine geodesic $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$ ($Dt^a/Du = 0$: tangent parallel-transported). General param: RHS = $(\ddot{s}/\dot{s})\dot{x}^a$; affine $\Leftrightarrow u = as + b$. **La-grangian method** (avoids computing Γ): vary $L = g_{ab}\dot{x}^a \dot{x}^b$ with Euler-Lagrange $\frac{d}{du}(\partial L/\partial \dot{x}^a) = \partial L/\partial x^a$, then read Γ off the equations. **First integral** $g_{ab}\dot{x}^a \dot{x}^b = \text{const}$ ($= c^2$ massive, 0 null). **Conserved component** if $\partial_c g_{ab} = 0$ then $t_c = g_{cb}\dot{x}^b = \text{const}$ (cyclic coordinate).

6. PARTICLE DYNAMICS

Minkowski $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, $\eta = \text{diag}(+, -, -, -)$, $\Gamma = 0$. Λ linear with $\eta_{\mu\nu} = \Lambda^\rho_\mu \Lambda^\sigma_\nu \eta_{\rho\sigma}$; proper: $\det \Lambda = 1$, $\Lambda^0_0 \geq 1$.

4-velocity, 4-acceleration

4-velocity $u^\mu = dx^\mu/d\tau = \gamma_u(c, \vec{u})$, $u \cdot u = c^2$, $dt/d\tau = \gamma_u$. Free massive particle $\Rightarrow$ timelike geodesic. **4-accel.** $a^\mu = Du^\mu/D\tau$, $a \cdot u = 0$, $|a|^2 = -|\vec{a}_{\text{IRF}}|^2$ (spacelike). Proper accel (ex.2 Q9): $\alpha^2 = \gamma_u^6 (\vec{u} \cdot \vec{a})^2 / c^2 + \gamma_u^4 \vec{a} \cdot \vec{a}$; circular r : $\alpha = \gamma_u^2 v^2/r$.

Energy & momentum

4-momentum $p = mu$, $p^\mu = (E/c, \vec{p})$, $E = \gamma_u mc^2$, $\vec{p} = \gamma_u m \vec{u}$; $|p|^2 = m^2 c^2 \Rightarrow E^2 - |\vec{p}|^2 c^2 = m^2 c^4$. **4-force** $f^\mu = Dp^\mu/D\tau = \gamma_u (\vec{f} \cdot \vec{u}/c, \vec{f})$, $f \cdot u = 0$. **Photon** $E = |\vec{p}|c$, p null, null geodesic, $p^\mu = dx^\mu/d\lambda$; $k^\mu = p^\mu/\hbar$.

Questions & measurement

Method conserve total p^μ ; use frame-invariance of $|p_{\text{tot}}|^2$. Compton: $\bar{\nu} = \nu/[1 + (h\nu/m_e c^2)(1 - \cos \theta)]$. Single-photon emission/absorption by a free particle is impossible (ex.3 Q3). Thresholds (ex.3 Q4): evaluate $|p_{\text{tot}}|^2$ in the CM frame. **Doppler (4-vec)** (ex.2 Q8): observers u, v give $u \cdot v = c^2 \gamma_V$ and $\nu_E/\nu_R = u^\mu p_\mu / v^\nu p_\nu$. **Tetrad** orthonormal

$e_\mu(\tau)$: $g(e_\mu, e_\nu) = \eta_{\mu\nu}$, $ce_0 = u$; measurements are tetrad components.
Rotating frame $ds^2 = [c^2 - \omega^2(x^2 + y^2)]dt^2 + 2\omega y dt dx - 2\omega x dt dy - dx^2 - dy^2 - dz^2 \Rightarrow$ centrifugal + Coriolis.

7. ELECTROMAGNETISM

Lorentz force $f_\mu = qF_{\mu\nu}u^\nu$, $F_{\mu\nu} = -F_{\nu\mu}$, $F_{0i} = E_i/c$, $F_{12} = -B_3$ etc.
Field eqns $\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$, $\partial_{[\sigma} F_{\mu\nu]} = 0$ ($\Leftrightarrow$ Maxwell); $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.
Invariant (ex.3 Q5): $c^2|\vec{B}|^2 - |\vec{E}|^2 \propto F_{\mu\nu}F^{\mu\nu}$ is Lorentz-invariant; transform $\vec{E}, \vec{B}$ via $F' = \Lambda F$.

8. SPACETIME CURVATURE

Riemann tensor

$$(\nabla_a \nabla_b - \nabla_b \nabla_a)v_c = R_{abc}{}^d v_d$$

$$R_{abc}{}^d = -\partial_a \Gamma_{bc}^d + \partial_b \Gamma_{ac}^d + \Gamma_{ac}^e \Gamma_{be}^d - \Gamma_{bc}^e \Gamma_{ae}^d$$

Flat $\Leftrightarrow R_{abc}{}^d = 0$. **Symmetries** $R_{abcd} = -R_{bacd} = -R_{abdc} = R_{cdab}$, $R_{[abc]d} = 0$. Indep. components = $N^2(N^2 - 1)/12$ (1 in 2D, 6 in 3D, 20 in 4D). **Bianchi** $\nabla_{[a} R_{bc]d}{}^e = 0$.

Ricci, Einstein tensor

Ricci $R_{ab} = R_{cab}{}^c = R_{ba}$; scalar $R = g^{ab}R_{ab}$. **Einstein tensor** $G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R$, divergence-free $\nabla^a G_{ab} = 0$ (contracted Bianchi). **Unit 2-sphere** (ex.3 Q8) only indep. comp $R_{\theta\phi\theta\phi} = \sin^2 \theta$.

Physical manifestations

Geodesic deviation $\frac{D^2 \xi^a}{Du^2} = R_{dbc}{}^a \dot{x}^b \dot{x}^c \xi^d$; spacetime (tidal): $\frac{D^2 \xi^\mu}{D\tau^2} = R_{\nu\alpha\beta}{}^\mu u^\alpha u^\beta \xi^\nu$. Newtonian limit $\ddot{\zeta}^i = -(\partial^2 \Phi / \partial x^i \partial x^j) \zeta^j$.

Curved-vs-flat test (ex.3 Q7): compute R from the metric; e.g. $ds^2 = y^2 dx^2 + x^2 dy^2$ flat, but $y dx^2 + x dy^2$ curved.

9. GRAVITATIONAL FIELD EQUATIONS

Perfect fluid $T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu - pg^{\mu\nu}$; rest frame $\text{diag}(\rho c^2, p, p, p)$; conserved $\nabla_\mu T^{\mu\nu} = 0$.

Einstein equation

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\kappa T_{\mu\nu}, \quad \kappa = 8\pi G/c^4 \Delta\phi \approx 2\pi\alpha = \frac{6\pi GM}{a(1-e^2)c^2} \text{ per orbit}$$

Alt form $R_{\mu\nu} = -\kappa(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T)$. Vacuum: $R_{\mu\nu} = 0$ (not necessarily flat). With cosmological constant: $G_{\mu\nu} + \Lambda g_{\mu\nu} = -\kappa T_{\mu\nu}$, $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$.

Weak-field limit

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $|h| \ll 1$; $g_{00} \approx 1 + 2\Phi/c^2$, $T_{00} \approx \rho c^2 \Rightarrow \nabla^2 \Phi = 4\pi G\rho$ (Poisson). With Λ : $\nabla^2 \Phi = 4\pi G\rho - \Lambda c^2$ (repulsion). Vacuum energy $T_{\mu\nu} = \rho c^2 g_{\mu\nu}$, $p = -\rho c^2$.

10. SCHWARZSCHILD SOLUTION

$$ds^2 = c^2 \left(1 - \frac{2\mu}{r}\right) dt^2 - \left(1 - \frac{2\mu}{r}\right)^{-1} dr^2 - r^2 d\Omega^2$$

$\mu = GM/c^2$; $r_s = 2\mu$. **Birkhoff** any spherically-symmetric vacuum solution is static & equals Schwarzschild.

Geodesics & orbits

Conserved (Lagrangian): $(1 - 2\mu/r)\dot{t} = k$ (energy), $r^2\dot{\phi} = h$ (ang. mom., plane $\theta = \pi/2$). First integral $= c^2$ (massive) / 0 (null). **Energy eqn** $\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2}c^2(k^2 - 1)$,

$$V_{\text{eff}} = -\frac{GM}{r} + \frac{h^2}{2r^2} \left(1 - \frac{2\mu}{r}\right) \text{ (massive)}$$

$$\text{Massless: } \frac{1}{2}\dot{r}^2 + \frac{h^2}{2r^2} \left(1 - \frac{2\mu}{r}\right) = \frac{1}{2}c^2 k^2.$$

Circular orbits $r_\pm = \frac{h^2}{2\mu c^2} (1 \pm \sqrt{1 - 12\mu^2 c^2 / h^2})$; exist iff $h > \sqrt{12}\mu c$. $r_- \in (3\mu, 6\mu)$ unstable, $r_+ > 6\mu$ stable. **ISCO** $r = 6\mu$, $h = \sqrt{12}\mu c$; binding $\approx 7\% = \sqrt{3}(3\sqrt{2} - 4)/6 \Rightarrow$ accretion power. **Photon sphere** $r = 3\mu$ (peak of massless V_{eff}). **Orbit eqn** ($u = 1/r$): massive $u'' + u - 3\mu u^2 = GM/h^2$; massless $u'' + u - 3\mu u^2 = 0$.

Redshift & symmetry

Grav. redshift satellite/clock (ex.3 Q6 weak; ex.4 Q2 full):

$$\frac{\Delta\tau_C}{\Delta\tau_{C'}} = \left(1 - \frac{3\mu}{r}\right)^{1/2} \left(1 - \frac{2\mu}{R}\right)^{-1/2} \rightarrow 1 - \frac{3GM}{2rc^2} + \frac{GM}{Rc^2} \text{ (just g)} \rightarrow \text{const; radiation } (p = \rho c^2/3) \rightarrow \text{const.}$$

Radial-fall redshift (ex.4 Q3): $z \rightarrow \infty$ as $r_e \rightarrow 2\mu$. **Killing vectors** (ex.4 Q1): $\nabla_a \xi_b + \nabla_b \xi_a = 0$; metric indep. of $x^0 \Rightarrow e_0$ Killing; $\xi_a t^a$ conserved on geodesics.

11. CLASSIC TESTS OF GR

Perihelion advance (from the $-3\mu u^2$ term):

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\kappa T_{\mu\nu}, \quad \kappa = 8\pi G/c^4 \Delta\phi \approx 2\pi\alpha = \frac{6\pi GM}{a(1-e^2)c^2} \text{ per orbit}$$

Mercury: $43''$ per century. **Light bending** solve $u'' + u = 3\mu u^2$ perturbatively from a straight line:

$$\Delta\phi = \frac{4GM}{c^2 b}; \quad \text{Sun } 1.75''$$

Apparent size (ex.4 Q4): grazing ray reaches ∞ with $b = r(1 - 2GM/c^2 r)^{-1/2}$ ($r > 3\mu$); Sun appears ~ 3 km larger.

12. SCHWARZSCHILD BLACK HOLES

Singularities metric singular at $r = 2\mu$ (coordinate, removable) and $r = 0$ (true). **Kretschmann** $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \propto \mu^2/r^6$: finite at 2μ , diverges at 0. **Horizon** $r = 2\mu$ is a one-way membrane; an infalling observer crosses in finite proper time but coordinate time $t \rightarrow \infty$. Once inside (ex.4 Q5), reaches $r = 0$ in proper time $\tau \leq \pi\mu/c$ unavoidably.

13. COSMOLOGY

FRW geometry

$$ds^2 = c^2 dt^2 - a^2(t)[d\chi^2 + S_K^2(\chi) d\Omega^2]$$

$S_K = \sin(\sqrt{K}\chi)/\sqrt{K}$ ($K > 0$), χ ($K = 0$), $\sinh(\sqrt{|K|}\chi)/\sqrt{|K|}$ ($K < 0$). **Hubble** $H(t) = \dot{a}/a$; proper dist $l = a\chi$. **Redshift** $1 + z = \lambda_R/\lambda_E = a(t_R)/a(t_E)$ (from $Ea = \text{const}$ for photons). **Massive particle** (ex.4 Q7): $\gamma_v v \propto 1/a(t)$; $v \rightarrow c$ recovers redshift.

Friedmann equations

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{Kc^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{1}{3}\Lambda c^2$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{1}{3}\Lambda c^2$$

Fluid eqn $\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p/c^2) = 0$: dust $\rho a^3 = \text{const}$; radiation ($p = \rho c^2/3$) $\rho a^4 = \text{const}$.

Cosmological models

Critical density $\rho_{\text{crit}} = 3H^2/8\pi G$; for $\Lambda = 0$, $\rho \geq \rho_{\text{crit}} \Leftrightarrow K \geq 0$ (closed/flat/open). $\Lambda = 0, p \geq 0 \Rightarrow \ddot{a} < 0$ (Big Bang), age $< 1/H_0$; $H_0 \approx 68$ km/s/Mpc. **Static models** (ex.4 Q8): no static solution with $\Lambda = 0, \rho > 0, p \geq 0$; Einstein static needs $\Lambda > 0$ but is unstable. **Milne universe** (ex.4 Q6): empty $\Lambda = 0$ solution $ds^2 = c^2 dt^2 - c^2 t^2 [d\chi^2 + \sinh^2 \chi d\Omega^2]$ is just Minkowski (open spatial sections).